

Lucian Mocan

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U.S. Permanent Resident | No sponsorship required

EXPERIENCE

Visiting Student Researcher

Feb. 2026 – Present

Northwestern University

On-site, Evanston, Illinois, USA

- Developed **Medovik**, a compatibility-first model for running unmodified x86-64 Linux applications inside **ARM TrustZone** on a ROCKPro64 while preserving their existing operating-system and dependency environments
- Built the prototype using **OP-TEE**, x86-64 emulation, and a lightweight WebAssembly runtime, enabling containerized Linux applications to execute without a complete TEE-specific port
- Profiled application workloads and replaced only selected hot functions with C++ punch-through handlers, reducing slowdowns from **80–218x** to **1.15–35x** while leaving the remaining application environment unchanged

Language and Tooling Research Intern

June 2025 – Sep. 2025

University of Strasbourg

Remote

- Implemented **control-flow analysis** for functions and nested calls in **Althread's Rust** compiler and VM
- Built a modular **import system** by analyzing import mechanics in Promela, TLA+, Python, Go, and C++
- Developed **Althread's web IDE** (TypeScript, WASM) with a virtual filesystem, package manager, and debugger
- Authored comprehensive documentation and API guides for new users (writeup)

Full-stack Web Engineering Intern

June 2024 – Sep. 2024

Emerson

Cernay, France

- Translated purchasing workflows into an Angular + Express.js solution, **saving 800+ hours** of manual work
- Automated quote requests and centralized data management for the purchasing department (writeup)

EDUCATION

University of Strasbourg

Sep. 2024 – Aug. 2026

M.S. in Computer Science – Networks, Internet and Systems Engineering

Illkirch-Graffenstaden, France

- Relevant Courses:** Advanced Algorithms, Compiler Design, OS Design, Inter- and Intra-Domain Routing, Cloud and Virtualisation, Distributed Algorithms, Embedded Programming, Advanced Cybersecurity, Wireless Networks

University of Strasbourg

Sep. 2021 – Aug. 2026

Cursus Master in Engineering (CMI): Internet, Systems and Networks – Selective 5-year track

Strasbourg, France

University of Strasbourg

Sep. 2021 – June 2024

B.S. in Computer Science

Strasbourg, France

- Honors:** Rank: 6/120 | **Courses:** Operating Systems Architecture, Algorithms & Data Structures, Software Engineering, Computer Networks, Artificial Intelligence, Parallel Computing

AWARDS & FUNDING

Thinking Machines Lab | Tinker Research Grant

2026

- Awarded **\$5,000 in compute credits** to support research on LLM-assisted code porting

TECHNICAL SKILLS

Programming: C/C++, Rust, Python, TypeScript, WebAssembly | **Systems:** Linux, Bash, Docker, containers, virtualization, pthreads, MPI, OpenMP, shared memory

Infrastructure/Web: Nomad, Consul, Git, React, Angular, React Native, Express.js, SQL/MySQL

Networking/Security: ARM TrustZone, TEEs, sandboxing, TCP/IP, BGP, OSPF, IS-IS, MPLS, P4, 6LoWPAN

PROJECTS

Sharing Square | Capstone Project, Team Lead

Sep. 2025 – Jan. 2026

- Led a 5-person team building a privacy-centric IoT platform for crowdsourcing health, environmental, and smart-city data for research (writeup)
- Built web (Next.js) / mobile (React Native) components and integrated backend services (Gin), database infrastructure (MongoDB), and LLM-assisted data interaction features.

6LoWPAN Fragment Forwarding | Contiki-NG, nRF52840-DK

Dec. 2025

- Extended Contiki-NG's 6LoWPAN stack with 6LFF fragment forwarding in embedded C on an nRF52840-DK
- Validated IPv6 fragment forwarding across a three-node nRF52840-DK testbed, testing forwarding state and end-to-end packet delivery (code)